Panskura Banamali College (Autonomous)

M.Sc. 2nd Semester Examination - 2021 (Online Mode)
Subject: Computer Science Paper: COS-202(M1)
Paper Name: Automata Theory

F.M.: 20 **Time : 1 Hour**

A. Answer any one question: **1×4=4**

1. What is the transition graph? With suitable example.
2. Write the relation between Regular language and Regular Expression.

B. Answer any two questions: **2×8=16**

1. Find the Deterministic Finite Automata(DFA)graph of following expression :
   - i. a+ where ∑(a)
   - ii. a* where ∑(a)
   - iii. Find odd number of 'a' where ∑(a,b).
   - iv. Find even number of 'a' where ∑(a,b)

2. Compare between regular language and regular expression. What is Push Down Automata(PDA)? What is non-regular language? 4+2+2

3. Write short note of following: 4 x 2.5
   - i. Kleene Closure.
   - ii. Pumping Lemma
   - iii. Application of Automata Theory
   - iv. Extended Transition function

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